

Can Agentic AI Settle a 20-Year-Old Open Problem?

The decidability of $\mathcal{ALCI}_{\text{RCC5}}$ — a machine-checked campaign

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July 2026

The one-slide version

- In 2002/2003 I defined a family of **spatial description logics** and left their central decidability question **open**. It still is.
- In 2026 I directed a **four-month AI campaign** on it: two frontier models, adversarial review, machine verification.
- The question is **not settled**. But it is now *reduced* — honestly and partly kernel-checked — to **one well-posed lemma**, with a decidable fragment, a certified normal form, a working reasoner, and an exact arithmetical address (Π_1^0) along the way.

How to read everything in this talk

The mathematics was produced by AI assistants and is *unverified by human experts*.

Standing label: **strongly supported, not certified** — except the parts checked by the **Lean 4 kernel**, which are machine-verified (but still human-unreviewed).

Five ways two regions can relate (RCC5)



DR

disjoint



PO

partial overlap



EQ

equal



PP

proper part



PPI

proper superpart

- Every pair of regions stands in **exactly one** of these five relations.
- *Jointly exhaustive, pairwise disjoint* — the classic qualitative spatial calculus (a coarser cousin of RCC8).
- Think GIS, layout, biology, any “what contains / touches what” domain — **no coordinates required**.

Reasoning without geometry: the composition table

If I know $x R y$ and $y S z$, what can $x ? z$ be?

known	known	forced / allowed
$x PP y$	$y PP z$	$x PP z$ (<i>forced</i> : parts of parts are parts)
$x PP y$	$y DR z$	$x DR z$ (<i>forced</i> : inside something disjoint)
$x DR y$	$y DR z$	anything at all (<i>free</i> : 5 options)

- All 25 cells form the **composition table** — pure algebra, derived once from set semantics, then used forever.
- **Abstract semantics**: a “model” is any relation assignment closed under this table — *space itself has left the building*.
- Every probe in this project re-derives the table from scratch, so the algebra can never silently drift.

Description logic in one slide

Concepts = definitions built from properties and **quantified relations**:

$\text{GermanCity} \equiv \text{City} \sqcap \exists \text{PP. Germany}$ “a city *properly inside* Germany”

$\text{CityAtRiver} \equiv \text{City} \sqcap \exists \text{PO. River}$ “a city *overlapping* some river”

A **reasoner** then derives the taxonomy *automatically*:

$\text{Hamburg} \sqsubseteq \text{GermanCity}$ and $\text{Hamburg} \sqsubseteq \text{CityAtRiver}$.

- The decision problem underneath everything: **concept satisfiability** — “can this definition be realized at all?”
- Subsumption, classification, consistency all reduce to it.

$\mathcal{ALCI}_{\text{RCC5}}$: the spatial relations *become* the roles

The logic (Wessel 2002/2003):

- $\mathcal{ALCI}$ = the basic DL with inverse roles;
- roles = the five RCC5 relations themselves;
- the composition table is enforced **globally**;
- EQ = identity (*strong* semantics).

So one can say things like

$\forall \text{PP}. \neg \text{Lake} \quad \exists \text{DR}. \text{Reserve}$

“no part is a lake”, “disjoint from some reserve”.

Why this is different

Concrete-domain DLs constrain spatial *attributes* of objects. Here the relations are *first-class roles*: one can **quantify over them**. None of the known concrete-domain decidability results apply.

Why the standard toolbox dies on contact

Every pair of elements carries a relation $\Rightarrow$ a model is a **complete labelled graph**, never a tree.

no tree models

Blocking/unravelling —
the workhorse of DL
decidability — has no tree
to work on.

no finite models

$\exists P P I . T \sqcap \forall P P I . \exists P P I . T$
forces an infinite ascending
part-of tower.

nondeterministic table

Most cells allow several
outcomes — composition
constrains but rarely
determines.

Creating one new element forces relations to every existing element — and a dormant $\forall$
anywhere may wake up because of it.

The landscape: open exactly at the top

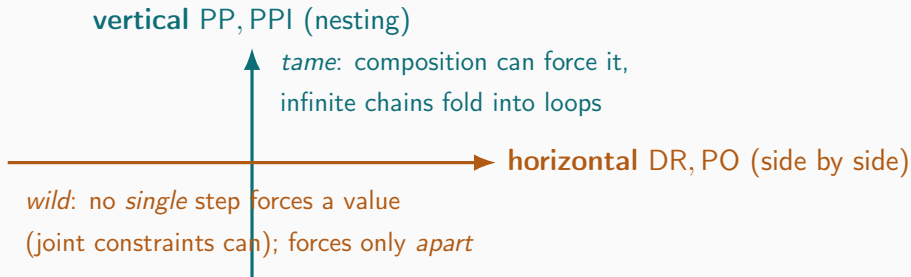
Logic	Relations	Concept satisfiability
$\mathcal{ALCI}_{\text{RCC1}}$	{SR}	decidable ($\equiv$ S5)
$\mathcal{ALCI}_{\text{RCC2}}$	{DR, O}	decidable
$\mathcal{ALCI}_{\text{RCC3}}$	{DR, ONE, EQ}	decidable (via two-variable FO!)
$\mathcal{ALCI}_{\text{RCC5}, \forall\text{PO-free}}$	RCC5 w/o $\forall\text{PO}$	decidable (this project)
$\mathcal{ALCI}_{\text{RCC5}}$	all five	open ; PSpace-hard
$\mathcal{ALCI}_{\text{RCC8}}$	eight RCC8 rel.	open ; ExpTime-hard

Lutz & Wolter 2006, after proving the *topological* cousins undecidable:

“Perhaps the most interesting candidate is $L_{\text{RCC5}}(\text{RS}) [= \mathcal{ALCI}_{\text{RCC5}}]$... to which the reduction exhibited in Section 8 does not apply.”

Recognized open by the leading experts — for **twenty years**.

One logic, two completely different axes



- Depth (nesting) is **never the problem** — ordinary modal-logic folding handles it.
- Everything hard lives in **horizontal crowds**: many pairwise-distinct neighbours that no finite summary seems to capture.

The asymmetry, made precise (machine-verified)

No *single* step forces a horizontal value. Of the 16 compositions of proper relations, exactly **four** return a single value:

$\text{comp}(\text{DR}, \text{PPI}) = \text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}, \quad \text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}, \quad \text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$

— every one runs through PP/PPI; no single cell yields $\{\text{PO}\}$. But the *intersection of several can force PO* ($\text{comp}(\text{DR}, \text{PO}) \cap \text{comp}(\text{PPI}, \text{PO}) = \{\text{PO}\}$) — so “shadow” is joint closure, not single-path.

Disjointness is half-deterministic.

downward: $x \text{ DR } y \Rightarrow$ every part of x is DR from every part of y

upward: a superpart of x may be DR, PO, or PPI to y — **free**

So one high DR edge dispatches an *arbitrarily wide* crowd of disjointnesses below it, for ^{9/37}

Shadows, live width, and the one property that matters

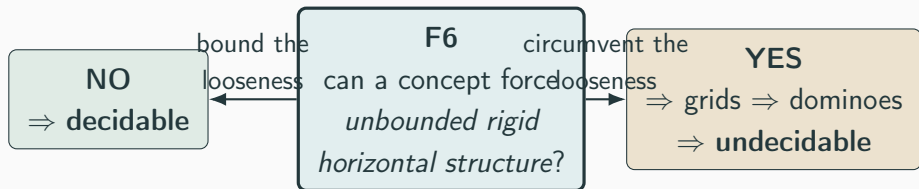
- An edge is a **shadow** if composition already forces it from edges you have — it costs a decision procedure *nothing*.
- The **live width** of an interface = the number of *non-shadow* edges there: the genuine branching that must be recorded.

Property F6 (the keystone of everything)

Is the live width of models bounded by a computable function of the input concept?

If yes $\Rightarrow$ finite certificate space $\Rightarrow$ **satisfiability is decidable**.

The knife's edge: one fact, two futures



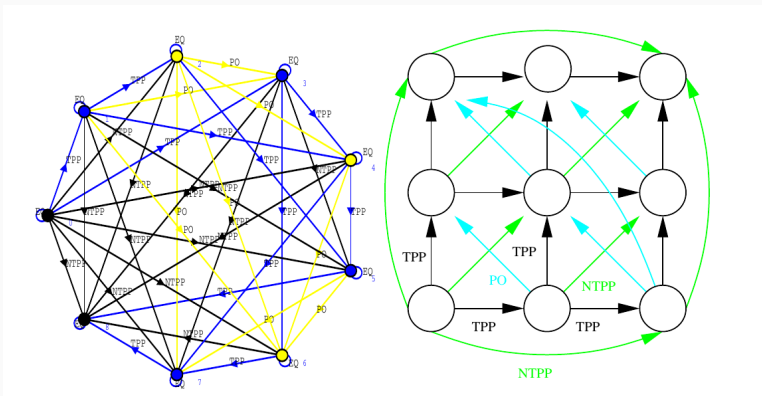
The *same* looseness of the composition table blocks **both** proofs. That is *why* neither direction has been settled in twenty years — it is one question seen from two sides.

The standard weapon needs a rigid grid

- Classic undecidability proofs encode **domino tilings** of $\mathbb{N} \times \mathbb{N}$: build a grid, force tile constraints, reduce the halting problem.
- The crux is **rigidity**: going *east then north* must land on *exactly the same cell* as *north then east* ($R_X \circ R_Y = R_Y \circ R_X$, the “coincidence condition”).
- Every known route into $\mathcal{ALCI}_{\text{RCC}}$ needs a feature it does not have: number restrictions, arbitrary role axioms, topological rigidity, nominals. . .

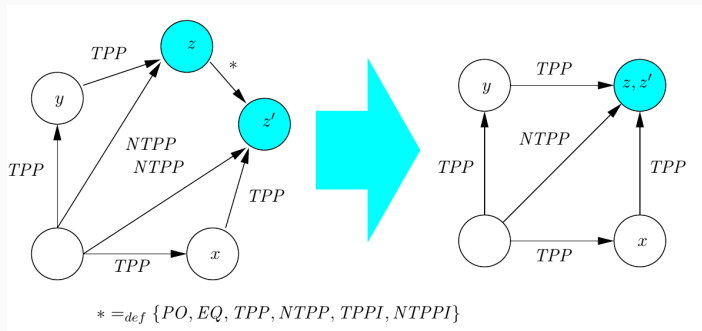
Question (2003 and today): can the composition table itself supply the rigidity?

2003: the grid models *exist*...



An RCC8 network isomorphic to an $n \times n$ grid — constructed by hand in my 2002/2003 report (report7, Fig. 10). The models are *out there*.

...but no concept can *force* them: the coincidence obstruction

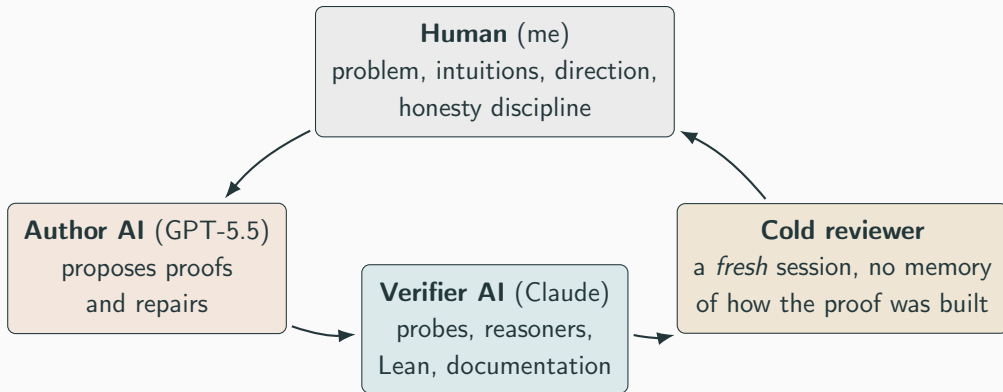


report7, Fig. 11: the two successors z (east-north) and z' (north-east) *should* be the same corner — but the table allows **six** relations between them. Leave it loose: the branches never join. Force a merge with isolation markers: *every* corner collapses to one point (verified again in 2026, at depth 2, for RCC5 and RCC8).

“Even though these infinite models do exist, we have found no way to enforce the depicted model(s) by means of a concept term.”

(report7. 2003)

The cast, and the loop



Propose → verify → **attack** → repair. Round after round, March–July 2026.

The campaign at a glance

- ~4 months, ~30 repair rounds
- 15 adversarial reviews — a defect or overclaim found in *all but two*
- 84 self-contained verification probes (each re-derives the algebra from set semantics)
- 4 Lean files, ~5,900 lines, zero sorry
- 2 working reasoners, cross-validated on 911 concepts

The single most telling number

Across *all* rounds and reviews, no accepted-but-unsatisfiable concept was ever found. Every defect was a gap in an *argument* — never a counterexample to the intended theorem.

That is why the label is *strongly supported* — and why it is still *not certified*.

Route 1: tableaux with blocking — broken by totality

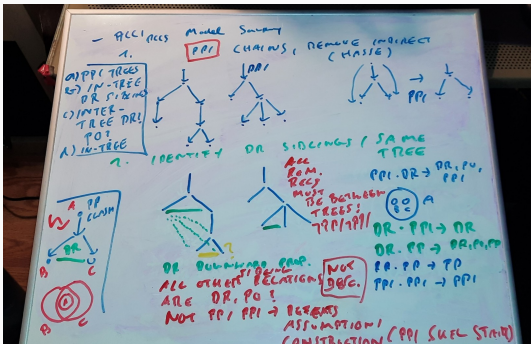
The default DL method: build a model top-down, *block* when a node repeats an ancestor's type.

- Here, composition forces edges **into already-built nodes**: from $x \text{ PP } y$, $y \text{ DR } z$ the table forces $x \text{ DR } z$.
- Diagnostic: $C_{\text{force}} = \exists \text{PP}.(\exists \text{DR}.A) \sqcap \forall \text{DR}.\neg A$ is **unsatisfiable** — any procedure that ignores forced edges wrongly accepts it.
- Dormant $\forall$'s reawaken blocked nodes; no fixed-arity local summary (types, triangles, quadruples, profiles, ...) is ever enough.

Lesson: in a total-relation logic, “local” does not exist.

What may be reused depends on the whole graph. (That is F6 in disguise.)

Route 2: the split-forest idea (a whiteboard, April 2026)



- **Present** the complete graph as a *forest of part-of trees*: split shared parts into EQ-mates, re-orient vertical edges into the trees.
- All remaining cross-edges are **horizontal**; complete them by the RCC5 *patchwork property* (a citable classical theorem).
- The one component **every** cold review found sound — the semantic foundation all later routes consume.

The human contribution: this decomposition. The models made it precise.

Route 3: tree automata — “acceptance without validity”

Package the search as a parity tree automaton (emptiness is decidable)...

- ...but the input tree **omits** the relations between fresh witnesses and the sources of $\forall$ -constraints.
- An *alternating* automaton explores in independent copies — and copies **guess the missing relations independently**.
- A cold review built the killer: two copies privately guess $\{PP\}$ and $\{DR\}$ for the same pair; $\text{comp}(PP, PP) \cap \text{comp}(PP, DR) = \{PP\} \cap \{DR\} = \emptyset$ — every copy accepts, **no model exists**.

So the automaton was **abandoned**, in favour of explicit finite certificates checked by composition tests — with a *truth lemma* instead of an acceptance condition. (This is what got formalized.)

Machine-checking: what the Lean kernel buys

- **Lean 4**: a proof assistant whose kernel type-checks *every* inference; a proof with a gap *does not compile*.
- Our discipline: Lean 4 **core only** — no libraries; every object built from first principles; tiny trusted base.
- “**zero sorry**” = no admitted gaps anywhere.

Why we moved to Lean

Thirteen of fifteen reviews broke a *prose* proof — typically at a step that was *assumed*, not proved. The kernel cannot be talked past. After the move, reviews stopped finding soundness defects and started finding only *interface* gaps — which were then closed.

What exactly is certified (zero sorry)

Result	Meaning	File
Soundness pipeline	valid certificate $\Rightarrow$ real RCC5 model	Round19Transport
Abstraction faithful	satisfiable $\Leftrightarrow$ Hintikka-realizable	Round19Transport
RCC5 normal form (fwd)	every network $\Rightarrow$ order + disjointness	RCC5NormalForm
Forcing reduction	F6 fails $\Leftrightarrow$ a concept forces crowds	ForcingReduction
Dovetail schema	qualitative F6 $\Rightarrow$ decidable	SemiDecidability

What is **deliberately not** in the kernel: the open completeness premise itself (F6). It is *stated*, never axiomatized — the honest boundary between what is proved and what is conjectured.

The main theorem (conditional, soundness certified)

Conditional decidability

If F6 holds — live width bounded by a computable function of the concept — then $\mathcal{ALCI}_{\text{RCC5}}$ concept satisfiability is **decidable**.

- The **soundness half** is kernel-checked end to end: certificate syntax $\rightarrow$ closure conditions $\rightarrow$ proper RCC5 frame $\rightarrow$ truth lemma $\rightarrow$ *actual model of the input concept*.
- The certificate checker is a real **executable Boolean function** — no oracle anywhere (a previous version had a hidden one; a cold review caught it; it was rebuilt).
- Everything that remains open is the **antecedent** — F6.

Unconditional win #1: a decidable fragment

Theorem ($\forall$ PO-free fragment)

Concepts with **no $\forall$ PO subformula** are decidable, with live width $\leq (1 + m)^2 2^{2^m}$,
 $m = |\text{cl}(C_0)|$.

- Still allowed: $\exists$ PO (“overlaps some. . .”), $\forall$ DR, $\exists$ DR, and **all** part-of modalities — a genuinely expressive spatial fragment.
- Why it works: PO is the *only* relation whose recurrence along a chain needs an “all-phase” safety condition; drop $\forall$ PO and the condition is vacuous.
- Found twice, independently, by two different routes — a good sign the fragment is natural, not an artifact.

Unconditional win #2: the RCC5 normal form

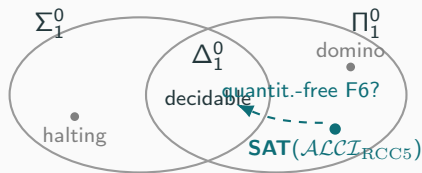
Theorem (forward direction kernel-checked, arbitrary domains; converse tested to size 4)

A strong-EQ RCC5 network is composition-closed *iff* it is an **ordered-disjoint structure** (the “ $\Rightarrow$ ” is the certified half):

- PP is a strict partial order (a genuine part-of hierarchy);
- DR is a symmetric, **downward-closed** disjointness;
- comparable pairs never disjoint; PO is the residual.

Every RCC5 model, anywhere, *is* just: a hierarchy + a downward-inherited “keeps-apart” relation + overlap as the leftover. The clean structural fact both major proof routes silently relied on — now a theorem with a 150-line proof the kernel accepts.

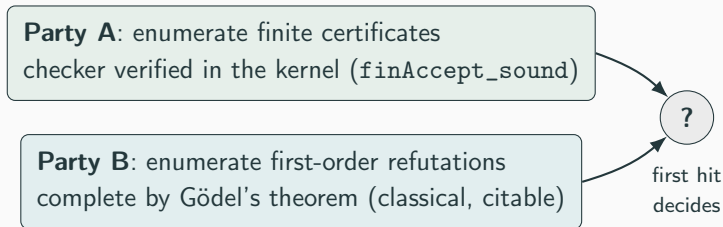
A late surprise: the problem's exact address is Π_1^0



- The whole semantics is **finitely first-order axiomatizable**; by Gödel completeness, *unsatisfiability* is enumerable.
- So SAT is **co-r.e.** (Π_1^0) — the domino problem's *ceiling* (membership; we don't claim hardness).
- Nobody in the project — or the reviews — had noticed for four months. Machine-cross-checked; the reduction is in Lean.

Consequence: half of the keystone was never needed

A Π_1^0 problem is decidable *iff* SAT is also enumerable. So run **two search parties** and wait for the first hit:



- Needed now: only **qualitative** F6 — *some* finite certificate per satisfiable concept, **no computable bound**.
- The bound the reviewers kept demanding (and that kept breaking) comes back **for free, afterwards** — a theorem of the schema.

“Isn’t this just first-order logic?” — yes, and it helps

$\mathcal{ALCI}_{\text{RCC5}}\text{-SAT}$ is FO satisfiability of a finite theory. Does it land in a **decidable fragment** of FO?

Fragment	Fate	Why
two-variable FO^2	×	composition is ternary (RCC3 <i>did</i> fall this way in 2003!)
prefix classes	×	unbounded $\forall\exists$ alternation, FMP
guarded / clique-guarded	×	totality is the one unguardable axiom
unary/guarded negation	×	finite model property; totality again
guarded + transitivity	×	undecidable outside guards

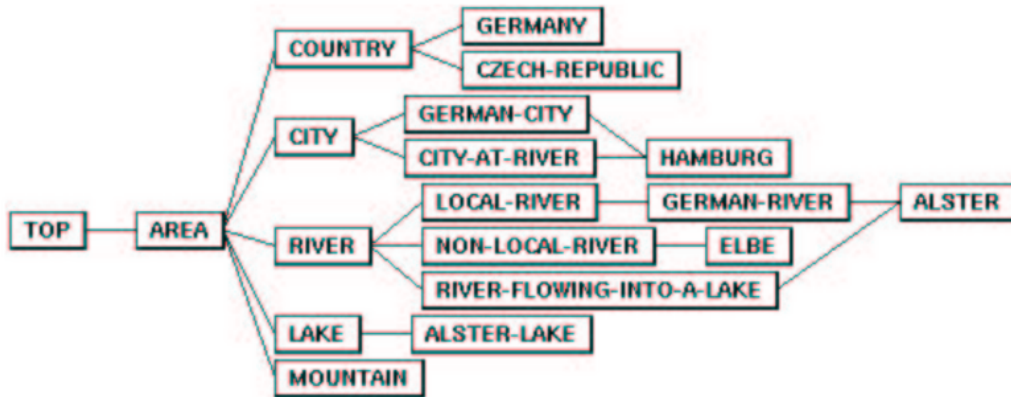
Every failure traces to the same fact: **every pair is related**, so models are maximally non-local — exactly the obstruction F6 names. The FO view rediscovers the project’s wall from classical model theory.

A working prototype reasoner (validated, not proved)

- The **cover-tree tableau** (~350 lines of Python): exact-occurrence blocking on the split-forest foundation.
- Cross-validated against an independent oracle on **911 concepts — zero mismatches**; 775/775 structured decomposition tests; never once contradicted in the whole campaign.
- Decides correctly even the known trap (the strong-EQ PO-loop) that fools type-based procedures.

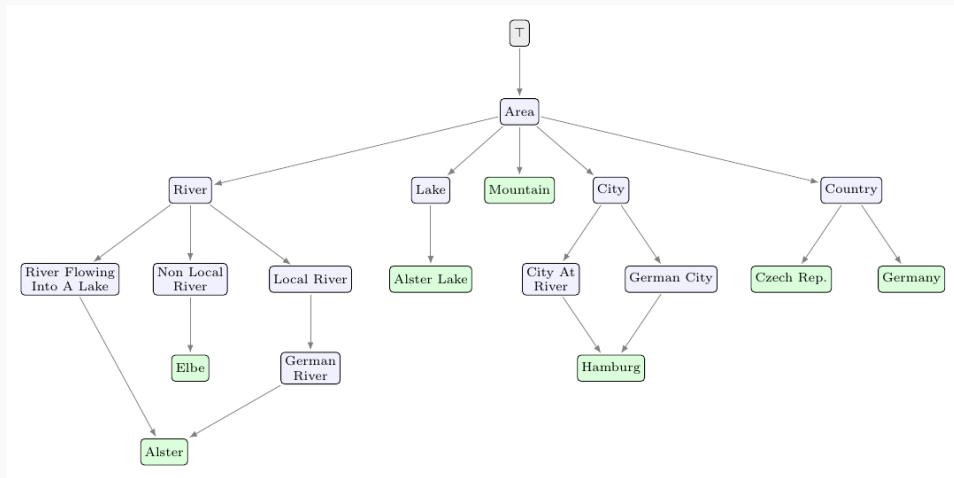
A running algorithm and a kernel-checked proof, **pointing the same way** — the strongest evidence short of a certificate.

2003, by hand



The GIS concept taxonomy I drew by hand in report7 (Fig. 6), 2003.

2026, by the reasoner: all 21/21 subsumptions — edge for edge



Recomputed automatically 23 years later (green = individuals; note *Hamburg* and *Alster* each have **two** parents — the DAG the reasoner must get right).

What the models did — and what they could not

Demonstrated

- months-long rigorous formal development, ~30 rounds
- genuinely useful *adversarial* review (with machine-checked witnesses)
- a 5,900-line kernel-checked Lean development
- working, validated software

Not demonstrated

- the **decisive novel idea** — F6 is uncracked
- reliable *self*-review (both models declared victory prematurely; both were caught)
- working unsupervised: the human directed every round

Neither model is self-sufficient. **What carried the project was the loop, not any single model.**

The method that worked (steal this)

1. **Cold review beats hot review.** A *fresh* session with no memory of the construction found what warm reviewers missed — every time it mattered. Make it a standard step.
2. **Don't trust — verify.** Every load-bearing claim was cross-checked mechanically: 84 probes that re-derive the algebra from set semantics, and ultimately the Lean kernel. The machine checks are the ground truth, **not the model's confidence**.
3. **Label honestly.** *Strongly supported* $\neq$ *certified*. Overclaiming happened twice; cold review caught both; the claims were withdrawn — **on the record**.

The full timestamped conversation log is public — every idea, defect, and repair traceable to its origin, human or model.

If nothing else: a message to fellow researchers

- Whatever the fate of the mathematics, this campaign is an **existence proof** of what frontier models could already do in serious formal research — *in July 2026*.
- Anyone doing serious logic or mathematics should **know** these capabilities exist. Ignoring them does not advance science; declining to use them is simply unwise.
- Saying exactly this was a stated purpose of the work. Its first outing (DL 2026) was rejected without engaging either the mathematics or the message — so it is now on arXiv, **in the open**, with the full audit trail.

The instruments have changed. The standards — verification, adversarial review, honest labels — matter more than ever.

One well-posed question
can a concept force unbounded
identity-selector structure?

NO $\Rightarrow$ decidable (machinery ready, kernel-checked) YES $\Rightarrow$ the grid/undecidability route opens

- 20+ years of difficulty, compressed into one lemma — with certified machinery waiting on **both** sides of the answer.
- Everything is frozen, tagged, reproducible: paper (37pp), 4 Lean files, 84 probes, 2 reasoners, full conversation log.

Open since 2003. Smaller than ever. Still standing.

`github.com/lambdamikel/alcircc5` — thank you!

Backup: the full RCC5 composition table

comp	$DR(a, b)$	$PO(a, b)$	$EQ(a, b)$	$PPI(a, b)$	$PP(a, b)$
$DR(b, c)$	*	DR, PO, PPI	DR	DR, PO, PPI	DR
$PO(b, c)$	DR, PO, PP	*	PO	PO, PPI	DR, PO, PP
$EQ(b, c)$	DR	PO	EQ	PPI	PP
$PP(b, c)$	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
$PPI(b, c)$	DR	DR, PO, PPI	PPI	PPI	*

Row = $S(b, c)$, column = $R(a, b)$, entry = possibilities for (a, c) ; * = all five. Re-derived from finite set semantics by every probe; matches the certified Lean table cell for cell.

Backup: what a finite certificate looks like

- **Catalogue** (the finite blueprint):
 - a *core network* — the non-repeating part of the model;
 - finitely many *attachment templates* — recipes for gluing a fresh witness cluster onto an interface;
 - a *steering function* fixing the horizontal relations from old occurrences to fresh ones (the only real freedom).
- **Plan**: which template to apply where; *unfolding* it (possibly forever) produces the split-forest model.
- **Soundness**, kernel-checked: one finite check on the catalogue guarantees *every* unfolding is a legal RCC5 model.
- **Live width** = live occurrences that must coexist at one interface during unfolding — the quantity F6 must bound.

Backup: the sharpened F6 — identity-selector minors

- The naive ways to force width (shell half-graphs, prefix pumps, PO-gaps) all **collapse** to finite catalogues — machine-checked.
- The sharp obstruction: n pairs (L_i, U_i) , each kept apart by a **private** selector A_i and by nothing else — the selector incidence matrix is the $n \times n$ *identity*.
- A finite catalogue can hold only boundedly many private selectors, so: **F6 fails iff some concept forces such minors for every n .**
- Bare RCC5 networks with unbounded such structure *exist* (machine-checked) — so any F6 proof must use $\mathcal{ALCI}_{\text{RCC}}$ -*forceability*, not RCC5 consistency alone. The uniformization side-condition (W2') provably folds into F6.